nRMSE (log colour scale; bold: best per domain)

	val	interp	new mesh	extrap	path	fine grid	large plate
GCN	0.022	0.022	0.023	0.039	0.130	0.022	0.068
A: MeshGraphNet	0.0036	0.0043	0.012	0.016	0.099	0.0054	0.011
B: + linear diffusion	**0.0029**	**0.0031**	0.0099	**0.0045**	0.076	**0.0046**	0.0092
C: enthalpy output	0.0049	0.0056	0.012	0.017	0.109	0.0066	0.011
D: conservative enthalpy	0.0062	0.0065	0.010	0.012	**0.012**	0.0072	**0.0084**
U-Net (grid)	0.0070	0.0074	**0.0074**	0.014	0.082	0.0075	0.350
LF-T (fitted filter)	0.340	0.344	0.345	0.303	0.345	0.345	0.347
Fixed linear filter	0.150	0.155	0.155	0.185	0.146	0.155	0.151

Melting-range MAE (K) (log colour scale; bold: best per domain)

	val	interp	new mesh	extrap	path	fine grid	large plate
GCN	13.2	17.3	17.7	42.1	326	17.2	11.9
A: MeshGraphNet	4.3	7.1	8.7	24.1	252	7.9	4.9
B: + linear diffusion	**3.7**	**3.9**	**4.7**	**4.8**	88.5	**5.2**	**4.6**
C: enthalpy output	4.2	6.8	8.2	24.5	277	8.2	4.8
D: conservative enthalpy	6.3	6.6	8.6	9.1	**11.5**	7.6	8.8
U-Net (grid)	8.1	8.7	8.8	14.5	79.5	9.0	340
LF-T (fitted filter)	146	150	155	161	235	151	157
Fixed linear filter	311	305	305	296	305	305	313